

Mode 12 — Complex

$$\text{C-DP4 (C16}\times\text{C16)} \rightarrow \text{DP8 (16}\times\text{16)} \rightarrow \text{DP8 (16}\times\text{8)} \rightarrow \text{DP8 (16}\times\text{4)} \rightarrow \text{DP8 (8}\times\text{4)}$$

Step 1 — C-DP4 (C16×C16)

$$\text{C-DP4 (C16}\times\text{C16)} = D = \sum_{i=0}^3 a_i b_i, \quad a_i = a_i^{re} + j a_i^{im}, \quad b_i = b_i^{re} + j b_i^{im}$$

Step 2 — DP8 (16×16)

$$\text{Re}(D) = \underbrace{\sum_{i=0}^3 a_i^{re} b_i^{re} - \sum_{i=0}^3 a_i^{im} b_i^{im}}_{\text{DP8 (16}\times\text{16)}}$$

$$\text{Im}(D) = \underbrace{\sum_{i=0}^3 a_i^{re} b_i^{im} + \sum_{i=0}^3 a_i^{im} b_i^{re}}_{\text{DP8 (16}\times\text{16)}}$$

Step 3 — DP8 (16×8) (byte-split $b = 2^8 b^{hi} + b^{lo}$)

$$\text{Re}(D) = \underbrace{2^8 \sum_{i=0}^3 a_i^{re} b_i^{re,hi} - 2^8 \sum_{i=0}^3 a_i^{im} b_i^{im,hi}}_{\text{DP8 (16}\times\text{8)}} + \underbrace{\sum_{i=0}^3 a_i^{re} b_i^{re,lo} - \sum_{i=0}^3 a_i^{im} b_i^{im,lo}}_{\text{DP8 (16}\times\text{8)}}$$

$$\text{Im}(D) = \underbrace{2^8 \sum_{i=0}^3 a_i^{re} b_i^{im,hi} + 2^8 \sum_{i=0}^3 a_i^{im} b_i^{re,hi}}_{\text{DP8 (16}\times\text{8)}} + \underbrace{\sum_{i=0}^3 a_i^{re} b_i^{im,lo} + \sum_{i=0}^3 a_i^{im} b_i^{re,lo}}_{\text{DP8 (16}\times\text{8)}}$$

Step 4 — DP8 (16×4) (nibble-split each byte)

$$\begin{aligned} \text{Re}(D) = & \underbrace{2^{12} \sum_{i=0}^3 a_i^{re} b_i^{re,hh} - 2^{12} \sum_{i=0}^3 a_i^{im} b_i^{im,hh}}_{\text{DP8 (16}\times\text{4)}} \\ & + \underbrace{2^8 \sum_{i=0}^3 a_i^{re} b_i^{re,hl} - 2^8 \sum_{i=0}^3 a_i^{im} b_i^{im,hl}}_{\text{DP8 (16}\times\text{4)}} \end{aligned}$$

$$\begin{aligned}
& + \underbrace{2^4 \sum_{i=0}^3 a_i^{re} b_i^{re, lh} - 2^4 \sum_{i=0}^3 a_i^{im} b_i^{im, lh}}_{\text{DP8 (16}\times\text{4)}} \\
& + \underbrace{\sum_{i=0}^3 a_i^{re} b_i^{re, ll} - \sum_{i=0}^3 a_i^{im} b_i^{im, ll}}_{\text{DP8 (16}\times\text{4)}} \\
\text{Im}(D) = & \underbrace{2^{12} \sum_{i=0}^3 a_i^{re} b_i^{im, hh} + 2^{12} \sum_{i=0}^3 a_i^{im} b_i^{re, hh}}_{\text{DP8 (16}\times\text{4)}} \\
& + \underbrace{2^8 \sum_{i=0}^3 a_i^{re} b_i^{im, hl} + 2^8 \sum_{i=0}^3 a_i^{im} b_i^{re, hl}}_{\text{DP8 (16}\times\text{4)}} \\
& + \underbrace{2^4 \sum_{i=0}^3 a_i^{re} b_i^{im, lh} + 2^4 \sum_{i=0}^3 a_i^{im} b_i^{re, lh}}_{\text{DP8 (16}\times\text{4)}} \\
& + \underbrace{\sum_{i=0}^3 a_i^{re} b_i^{im, ll} + \sum_{i=0}^3 a_i^{im} b_i^{re, ll}}_{\text{DP8 (16}\times\text{4)}}
\end{aligned}$$

Step 5 — DP8 (8×4) (byte-split $a = 2^8 a^{hi} + a^{lo}$)

$$\begin{aligned}
\text{Re}(D) = & \underbrace{2^{20} \sum_{i=0}^3 a_i^{re, hi} b_i^{re, hh} - 2^{20} \sum_{i=0}^3 a_i^{im, hi} b_i^{im, hh}}_{\text{DP8 (8}\times\text{4)}} \\
& + \underbrace{2^{12} \sum_{i=0}^3 a_i^{re, lo} b_i^{re, hh} - 2^{12} \sum_{i=0}^3 a_i^{im, lo} b_i^{im, hh}}_{\text{DP8 (8}\times\text{4)}} \\
& + \underbrace{2^{16} \sum_{i=0}^3 a_i^{re, hi} b_i^{re, hl} - 2^{16} \sum_{i=0}^3 a_i^{im, hi} b_i^{im, hl}}_{\text{DP8 (8}\times\text{4)}}
\end{aligned}$$

$$\begin{aligned}
& + \underbrace{2^8 \sum_{i=0}^3 a_i^{re,lo} b_i^{re,hl} - 2^8 \sum_{i=0}^3 a_i^{im,lo} b_i^{im,hl}}_{\text{DP8 (8}\times\text{4)}} \\
& + \underbrace{2^{12} \sum_{i=0}^3 a_i^{re,hi} b_i^{re,lh} - 2^{12} \sum_{i=0}^3 a_i^{im,hi} b_i^{im,lh}}_{\text{DP8 (8}\times\text{4)}} \\
& + \underbrace{2^4 \sum_{i=0}^3 a_i^{re,lo} b_i^{re,lh} - 2^4 \sum_{i=0}^3 a_i^{im,lo} b_i^{im,lh}}_{\text{DP8 (8}\times\text{4)}} \\
& + \underbrace{2^8 \sum_{i=0}^3 a_i^{re,hi} b_i^{re,ll} - 2^8 \sum_{i=0}^3 a_i^{im,hi} b_i^{im,ll}}_{\text{DP8 (8}\times\text{4)}} \\
& + \underbrace{\sum_{i=0}^3 a_i^{re,lo} b_i^{re,ll} - \sum_{i=0}^3 a_i^{im,lo} b_i^{im,ll}}_{\text{DP8 (8}\times\text{4)}} \\
\text{Im}(D) = & \underbrace{2^{20} \sum_{i=0}^3 a_i^{re,hi} b_i^{im,hh} + 2^{20} \sum_{i=0}^3 a_i^{im,hi} b_i^{re,hh}}_{\text{DP8 (8}\times\text{4)}} \\
& + \underbrace{2^{12} \sum_{i=0}^3 a_i^{re,lo} b_i^{im,hh} + 2^{12} \sum_{i=0}^3 a_i^{im,lo} b_i^{re,hh}}_{\text{DP8 (8}\times\text{4)}} \\
& + \underbrace{2^{16} \sum_{i=0}^3 a_i^{re,hi} b_i^{im,hl} + 2^{16} \sum_{i=0}^3 a_i^{im,hi} b_i^{re,hl}}_{\text{DP8 (8}\times\text{4)}} \\
& + \underbrace{2^8 \sum_{i=0}^3 a_i^{re,lo} b_i^{im,hl} + 2^8 \sum_{i=0}^3 a_i^{im,lo} b_i^{re,hl}}_{\text{DP8 (8}\times\text{4)}}
\end{aligned}$$

$$\begin{aligned}
& + \underbrace{2^{12} \sum_{i=0}^3 a_i^{re,hi} b_i^{im,lh} + 2^{12} \sum_{i=0}^3 a_i^{im,hi} b_i^{re,lh}}_{\text{DP8 (8}\times\text{4)}} \\
& + \underbrace{2^4 \sum_{i=0}^3 a_i^{re,lo} b_i^{im,lh} + 2^4 \sum_{i=0}^3 a_i^{im,lo} b_i^{re,lh}}_{\text{DP8 (8}\times\text{4)}} \\
& + \underbrace{2^8 \sum_{i=0}^3 a_i^{re,hi} b_i^{im,ll} + 2^8 \sum_{i=0}^3 a_i^{im,hi} b_i^{re,ll}}_{\text{DP8 (8}\times\text{4)}} \\
& + \underbrace{\sum_{i=0}^3 a_i^{re,lo} b_i^{im,ll} + \sum_{i=0}^3 a_i^{im,lo} b_i^{re,ll}}_{\text{DP8 (8}\times\text{4)}}
\end{aligned}$$